Mandatory

Problem 1: Z-Score Method (Normal Data)

Dataset (Age of students in a class):

```
[18, 19, 20, 20, 21, 21, 22, 22, 23, 24, 25, 26, 27, 28, 30, 32, 35, 40, 45, 70]
```

Tasks:

- Calculate the mean and standard deviation of Age.
- Compute Z-Score for each value.
- Identify outliers using |Z| > 3 conditions.
- Remove those outliers and show the new dataset length.

Problem 2: IQR Method (Skewed Data)

Dataset (Monthly salaries in thousand BDT):

```
[25, 28, 30, 32, 32, 33, 35, 35, 36, 38, 40, 42, 45, 48, 50, 55, 60, 65, 70, 80,
120, 150, 200, 250, 500]
```

Tasks:

- Calculate Q1 (25th percentile) and Q3 (75th percentile).
- Find IQR = Q3 - Q1.
- Calculate lower bound and upper bound.
- Identify outliers.
- Cap (clip) the outliers to the bounds instead of deleting them.
- Show min and max after capping.

Problem 3: Choose the Right Method

Dataset A (Exam scores):

```
[55, 58, 60, 62, 65, 65, 68, 70, 72, 75, 78, 80, 85, 90, 95, 98, 100, 100, 100, 15,
20]
```

Note: There are two very low scores (15, 20) and three high scores (100).

Dataset B (House prices in lakh BDT):

```
[30, 32, 35, 38, 40, 42, 45, 48, 50, 55, 60, 65, 70, 80, 90, 100, 120, 150, 200,
300, 400, 500, 800, 1000, 1200]
```

Tasks:

- For Dataset A, plot a histogram (conceptually) – is it normal or skewed?
- Which method (Z-Score or IQR) would you choose for Dataset A? Why?
- For Dataset B, which method would you choose? Why?
- Apply IQR on Dataset B and count how many outliers you get.

Problem 4: Winsorization (Percentile Method)

Dataset (Product ratings out of 10):

```
[1, 2, 3, 4, 4, 5, 5, 5, 6, 6, 7, 7, 8, 8, 9, 9, 10, 10, 1, 2, 10, 10, 10, 0.5, 9.5]
```

Tasks:

- Use 5% Winsorization (x = 5, meaning cap at 5th and 95th percentiles).
- Calculate 5th percentile and 95th percentile.
- Cap all values below 5th percentile to the 5th percentile value.

- Cap all values above 95th percentile to the 95th percentile value.
- Show the min and max after Winsorization.
- Explain why Winsorization is better than deletion here.

Extra (Optional)

Problem 5: Mixed Methods – Compare Z-Score & IQR on Same Data

Dataset (Product ratings out of 10):

```
[12, 15, 18, 20, 22, 22, 23, 24, 25, 25, 26, 27, 28, 30, 32, 35, 38, 40, 45, 50, 55,
60, 65, 70, 80, 90, 100, 120, 150, 200, 500, 1000, 2000]
```

Tasks:

- Plot a histogram (conceptually) and decide if the data is normal or skewed.
- Apply Z-Score method (|Z| > 3) – count outliers and list them.
- Apply IQR method (1.5×IQR) – count outliers and list them.
- Which method found more outliers? Why?
- Which method would you trust here? Justify.

Problem 6: Real-World Scenario – Employee Bonus

Scenario:

A company gives an annual bonus (in thousand BDT). HR wants to remove extreme outliers before calculating average bonus for policy making.

Dataset (Bonus amounts):

```
[5, 5, 6, 6, 7, 7, 7, 8, 8, 8, 9, 9, 10, 10, 10, 11, 12, 13, 15, 18, 20, 25, 30, 35,
40, 50, 60, 80, 100, 120, 150, 200, 300]
```

Tasks:

- Calculate mean, median, and mode. What do you observe?
- Apply IQR method and find outliers.
- Instead of deleting, apply Winsorization with x = 3 (i.e., cap at 3rd and 97th percentile).
- Show the new min and max after Winsorization.
- Calculate the new mean after Winsorization. Is it more reasonable than the original mean?